

OPERATION MANUAL

Thermal Inkjet Printer Godavari



INSTALLATION AND USER MANUAL

Thermal Inkjet-Godavari

(With reference to Software Rel. 2.06
Updated November, 2019)

used on
High resolution TIJ printer

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WARNING: SYSTEM IS SUPPLIED WITH THE 110-220Vac POWER SUPPLY ALWAYS, DISCONNECT THE POWER SUPPLY CABLE PRIOR TO
ACCESSING ANY INTERNAL PARTS.

(Important note: The specifics and functioning of the system may vary in future versions.)

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1. DESCRIPTION OF THE SYSTEM

1.1 GENERAL DESCRIPTION OF THE SYSTEM

TIJ-Godavari are high definition printers for direct printing onto packaging materials which utilizes the Thermal Inkjet Technology (TIJ).

It is based on a print head with 300 vertical dots in two columns of 150 dots, individually addressable, which cover a height of 12.7 mm (0.5"). This allows for a vertical printing resolution of 600 dpi (dots per inch) when printing with both columns or 300 dpi when printing with one column only (ink economy).

The printhead is integrated into an ink cartridge which represents the ink system.

TIJ-Mini allows for the creation of images with a height of 300 dots (12.7 mm) and a width of about 5000 dots (about 800 mm when using a resolution identical to the vertical one. Aspect Ratio 1:1).

The printing of the vertical columns combined with the horizontal movement of the object to be printed on composes the complete image.

The various available matrix fonts can be mixed together to form the printing image: 5x5, 7x5, 9x7, 15x10, 16x10, 32x18.

To be readable, each dot of the matrix font is really formed by 8 dots in vertical and it is called a "macrodot".

In reality, the final image height is thus composed of 32 microdots'.

For instance, the biggest font available, 32x18, is 256 dot high (about 10.8 mm), while the smallest 5x5 is 40 dot high (about 1.7 mm).

The system consists of:

- 1) 7" touch screen display terminal with stylus and connecting cable is Interactive, easy to use, icon based, WYSIWYG graphical interface. The user friendly interface allows the preparation of the images to be printed and the setting of the various printing parameters in a simple and easy manner with minimal operator training.
- 2) Ink cartridge integrated into print head. Cartridge replacement leads to de facto replacement of the print head and ink delivery system. No print head cleaning, filter replacements, mechanical pumps replacement or other maintenance is required.
- 3) Controller for print head with smart card reader, input / output connectors integrated product sensor. LAN, encoder, external photocell and BCD inputs interfaces. External input possibilities allow for the control of diverse automated printing functions.

- 4) Industrial power supply with universal AC input.
- 5) Mounting brackets and accessories. Spring-loaded adapter for continued maintenance of a minimum distance between print head and substrate (Optional). Top and side print brackets and floor stand.

1.2 DISPLAY TERMINAL - USER INTERFACE

Terminal is equipped with a 7" TFT LCD 800x480 backlit type display with touch screen that can be used with or without a stylus. A QWERTY keyboard is shown on the display when data should be entered.

1.3 INK CARTRIDGE

- The ink delivery system and the print head are completely integrated into an autonomous cartridge.
- Ink capacity of the cartridge is 45 ml in the TIJ-Godavari. Upon switching on the system, no start-up procedure is required, the print head is ready for use.
- Neither is a shutdown procedure is required. Only in case of long shutdown, the cartridge should be removed from the printhead and closed with its clip.
- There are no replaceable filters in the ink path.
- Print head cleaning could be required just in case of dusty environment. A simple purging and wiping will recover the functionality of the print.
- Priming operation is activated by pressing the PURGE pushbutton on the touch screen.
- There is no ink level sensor. The remaining ink availability is managed by the smart card which is included with the cartridge.
- Each new cartridge comes with an ink credit (number of printable dots) which is stored into the smart card.
- Different cartridge types, depending on the type of ink, it may require different settings (temperature, pulse width, voltage).
- The accompanying smart card contains the ideal setting for that particular cartridge, using that setting the cartridge will last longer and will generate the best print results.

1.4 GRAPHICS – HYPERTEXTS

The printer allows for the management of texts, logos, bar codes, date and time codes, expiry dates, counters, shifts, batch, in any combination.

This combination is called hypertext.

FIXED TEXTS

Fixed texts may be added. Font dimension and position in x and y coordinates can be defined.

LOGO

Logos can be managed with their position and dimension.

VARIABLE TEXTS

Aside from fixed texts, the system may also be set to automatically generate variable texts.

- Counters to identify the number of prints generated, either incrementing or decrementing, numbers or letters, for production lot counts.
- Data to print day, month, year, week, sequential day in any format
- An expiry set in days, months or years in any format
- A time in hours and minutes
- A shift identifiable in 4 daily intervals and associated to a code, a name, a letter, a number.
- A batch / box code made up of fixed and variable part.

BARCODES

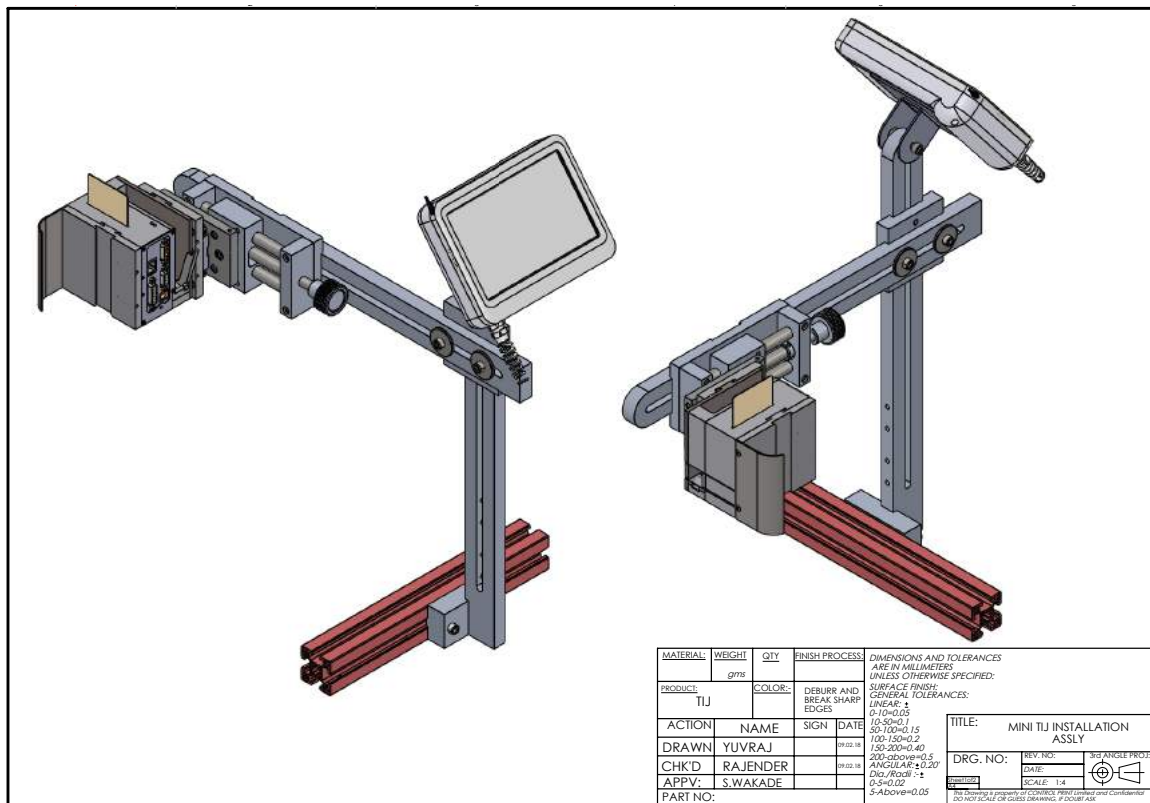
Barcodes of varying numeric or alphanumeric content type may be inserted into each hypertext. Each single bar code may be completely personalized. If the barcode is not used for internal handling purposes, the rules and standards relative to that type of code are used.

DIVISION OF THE IMAGE

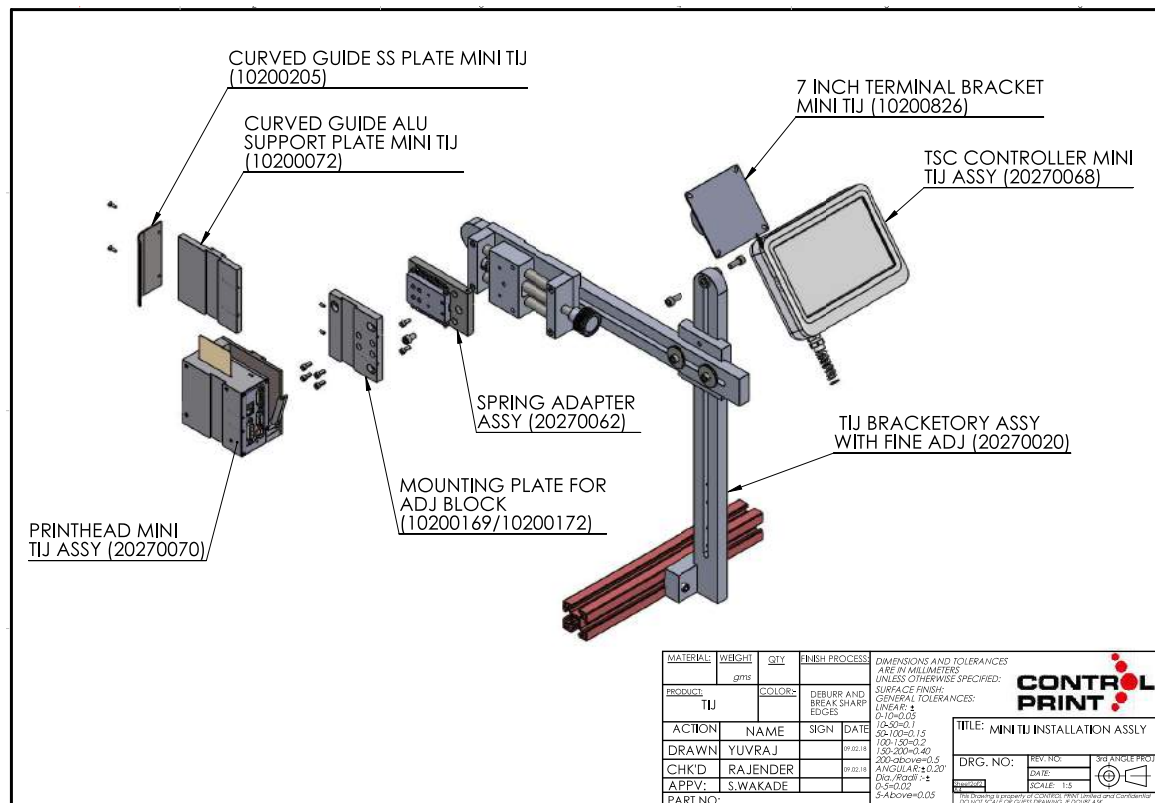
By selecting the number of columns to be printed it is possible to print only a portion of the maximum image allowed by the system (32x5000 dots). The bigger the image, the greater is the time for the system to prepare and print it. This process is known as codification and may take several seconds depending on the complexity of the image.

2. SYSTEM INSTALLATION

2.1 PRINTER INSTALLATION ON CONVEYOR



PRINTER INSTALLATION ON CONVEYOR: EXPLODED VIEW



2.2 PRINT HEAD AND INK DELIVERY SYSTEM

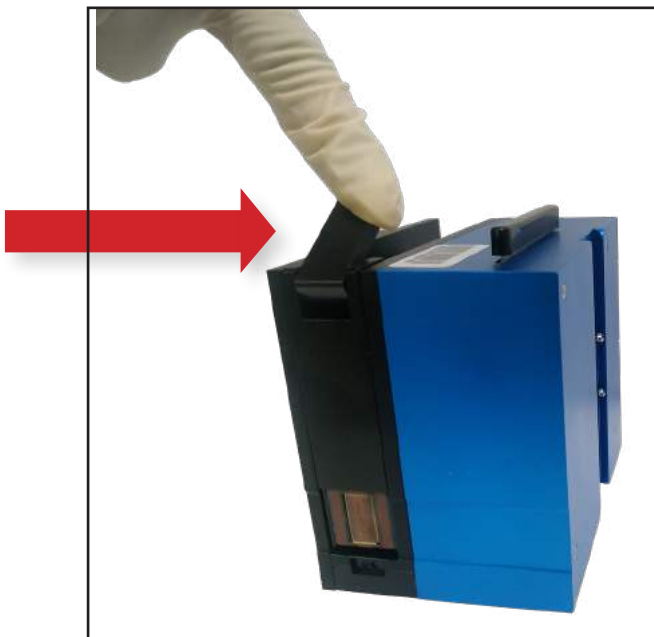
- The print head and the ink delivery system are integrated into a single cartridge.
- No tubes, filters, pumps or separate ink tank are present.
- The cartridge is plugged into the printhead which can be mounted to print on the side or on the top of the object to be marked.
- The printhead should be installed on a horizontal bar mounted on a vertical one (floor stand/fine adjustment bracketory).
- Install the vertical and horizontal bars in the required position.
- Allow for the printhead to move forwards and backwards so that it may be withdrawn when not in use.
- The printhead can be fitted with the spring-loaded adapter which allows the print head to maintain continuously the closest possible distance to the product to be printed on and asorbs mechanical shocks.
- As the printhead is installed in its final position, connect the cable to the terminal and the cable to the power supply.
- Then switch on the system by connecting the power supply to the mains.
- As the system starts up, the main display page is shown on the terminal.



- The red LED on the printhead will light to indicate that the system requires the insertion of the cartridge.
- Insert the cartridge until the nose (in the print head) exits appears through the front:



- Then lock it in place using the lever:
As the lever is up, in it's locked position, the printhead is switched on.



- Insert the smart card that came with the cartridge into its slot.
- This will automatically set the optimum cartridge operating parameters or user can use parameters specified on to cartridge label manually.
- The ink credit will now be shown on the main display page and the printhead is ready to print.

2.3 PRINthead CONNECTIONS

The printhead is provided with various connections to allow it to interface with the outside world.

On the rear of the printhead, the following connections are available (see on the side):

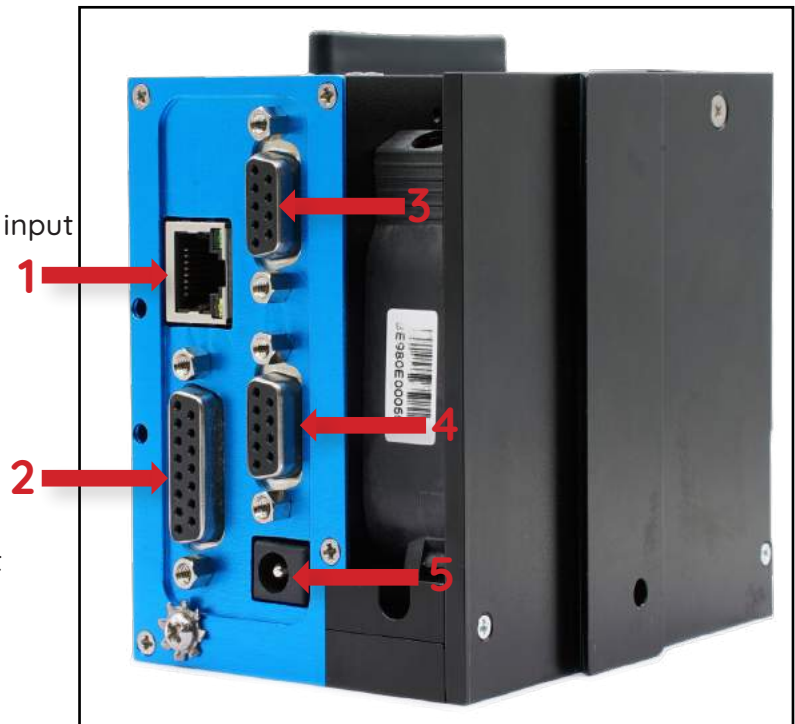
1) 8 pin RJ45 shielded plug for Ethernet 10/100 Mbps.

The printhead can be connected to other devices through Ethernet with TCP/IP protocol (with selected models only).

The system can be set and the hypertexts can be modified by sending messages through Ethernet. (with selected models only).

2) 15 pin female connector for BCD (Binary Coded Decimal) inputs and alarm relay's output.

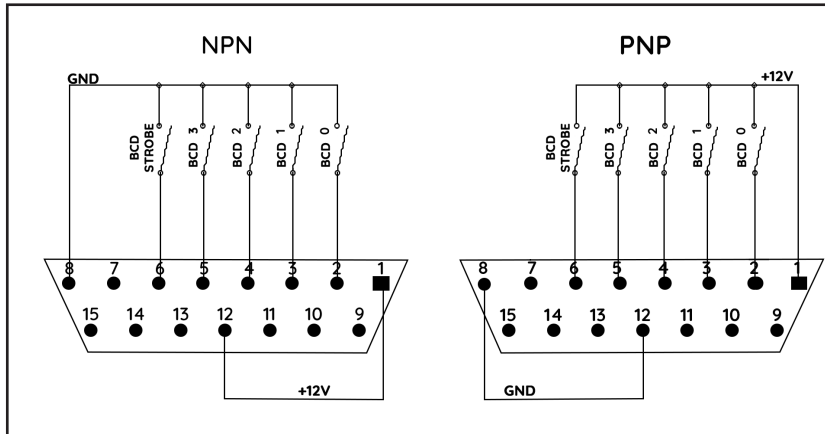
- pin 1: +15V
- pin 2: BCD 0 signal input
- pin 3: BCD 1 signal input
- pin 4: BCD 2 signal input
- pin 5: BCD 3 signal input
- pin 6: BCD Strobe signal input
- pin 7: NC relay out
- pin 8: GND
- pin 9: +5V
- pin 10: +5V
- pin 11: +15V
- pin 12: BCD Common
- pin 13: COM for relay out
- pin 14: NO for relay out
- pin 15: GND



BCD inputs

BCD inputs can be used to automatically select the hypertext to be printed by means of external signals coming from any plc server.

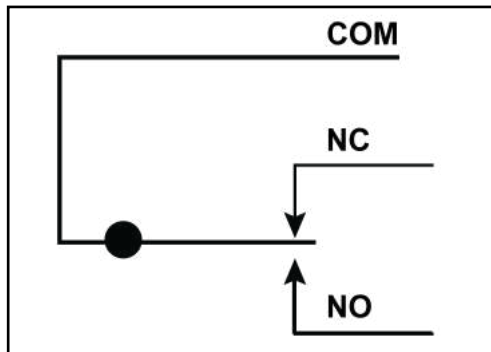
When using BCD inputs, based on the type of signal, NPN or PNP respectively connect the common to +12V (+15V) or to GND.



Relay output

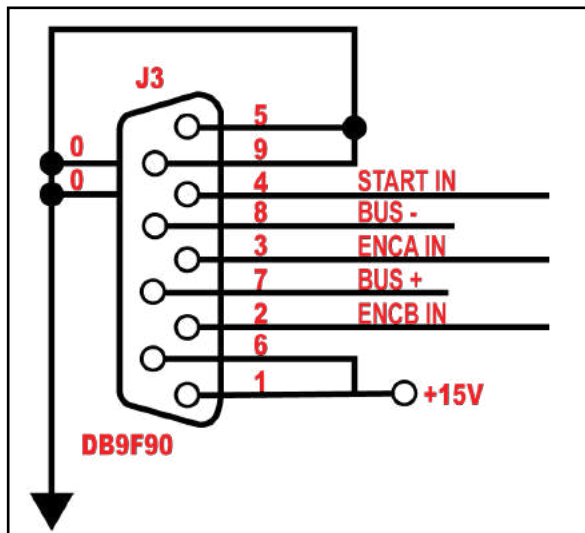
Relay output is used as an alarm.

It is a voltage-free contact:



3) 9 pole female connector (J3) for a second printhead or other devices.

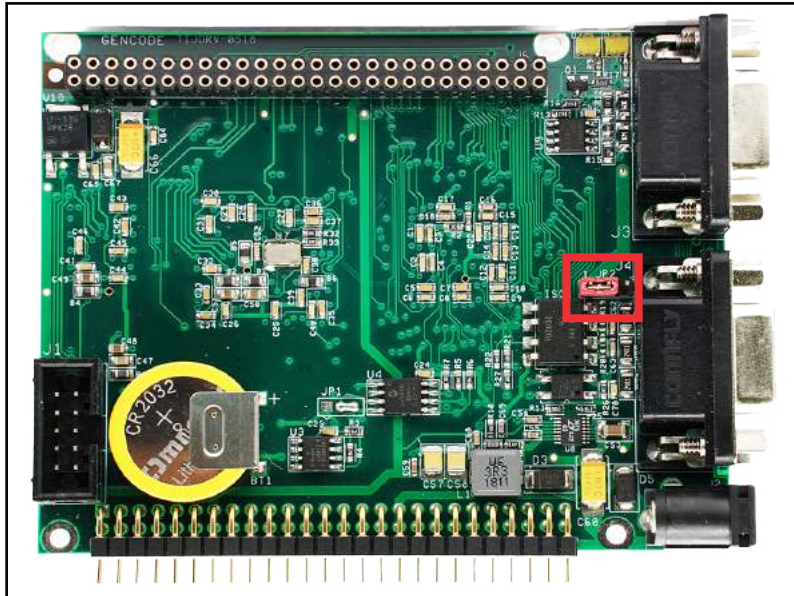
- pin 1: +15V
- pin 2: Encoder B output signal
- pin 3: Encoder A output signal
- pin 4: External photocell output signal
- pin 5: GND
- pin 6: +15V
- pin 7: +RS485
- pin 8: -RS485
- pin 9: GND



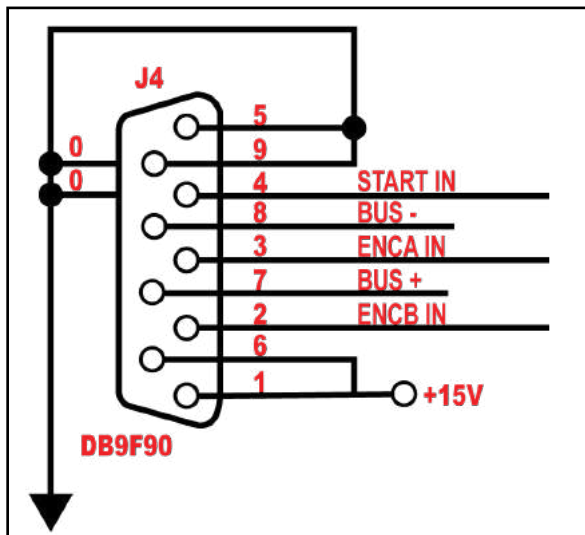
The external photocell can be PNP or NPN type.

- If a PNP is used, then the internal jumper JP2 should be set to 2-3.
- If a NPN is used, then the internal jumper JP2 should be set to 1-2. This is the default configuration.

(Highlighted in below snapshot with default position)



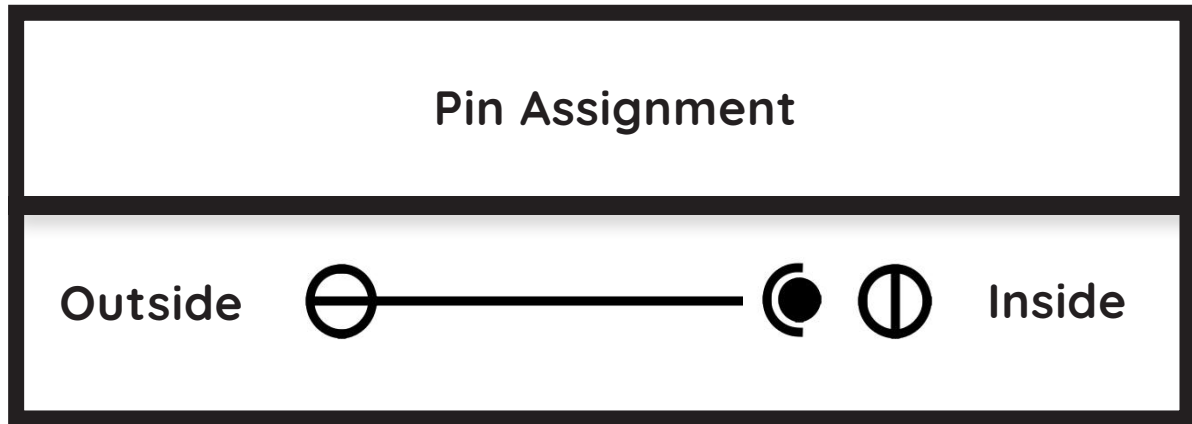
4) 9 pole female connector (J4) for the display terminal, an encoder, an external photocell (product sensor).



- pin 1: +15V
- pin 2: Encoder B input signal
- pin 3: Encoder A input signal
- pin 4: External photocell input signal
- pin 5: GND
- pin 6: +15V
- pin 7: +RS485
- pin 8: -RS485
- pin 9: GND

The encoder should be PNP type with A and B signals.
Connector J3 and J4 are identical in all manners.

5) Jack type socket for power supply connection.
External: +15V
Internal: GND



The mating plug should be of the following type:

- jack insertion depth: 9.0 mm
- External diameter: 5.5 mm
- Internal diameter: 2.5 mm

2.4 OPERATOR TERMINAL CONNECTION

Install the terminal at a level that allows the display to be clearly visible.
Floor stand, brackets and rear magnet are available for that purpose.

Connect the terminal to the printhead before switching on the power.

The 9 pole male connector has the following pin layout:

- pin 1: +15V
- pin 2:
- pin 3:
- pin 4: NOT TO BE USED (to be connected to an 8V for other applications)
- pin 5: GND
- pin 6
- pin 7: +RS485
- pin 8: -RS485
- pin 9: NOT TO BE USED (to be connected to an ON button for other application's)

A USB port type A and a **smart card reader** are also present on the front side of the terminal.

The terminal can be mounted by using the five M4 inserts and the 7" terminal bracket provided with the printer itself, on the back.

2.5 CONNECTION OF THE MAINS

A power supply is provided to generate the 15VDc required by the system.

Connect power supply to **100-240VAC, 50-60Hz ensuring the presence of a quality ground connection.**

Warning: if the ground conductor is not present, the system will not operate properly.

Furthermore, active and passive electrical safety protection would not be provided.

3. DATA

3

- All the data used by the system are stored into the internal memory of the terminal. Only the information required for the image to be printed (selected hypertext) is stored into the printhead.
- When the automatic selection of the message to be printed is enabled (in conjunction with BCD inputs), then 16 hypertexts are stored into the controller and are immediately ready to be printed as the BCD input configuration changes.
- All the data related to time and date codes are stored in the internal Real Time Clock (RTC) in the controller which is battery buffered to maintain that information during switch-off.
- With the data stored into the system it is possible to create printable images, modify the print settings, produce print reports, change the messages shown on the display customize the operation of the system.
- These data represent the archive which allows the system to operate.
- This is an extremely important archive and it is advisable to keep an updated copy.
- Data can also be stored on an USB memory key for backup.
- Data may be modified or read on line by connecting the printer to Ethernet and using a computer which is also connected to the net. (Available in future versions)

4. PROGRAMMING AND USING THE SYSTEM

4

The hypertexts and all other setting are programmed using icon based screens.

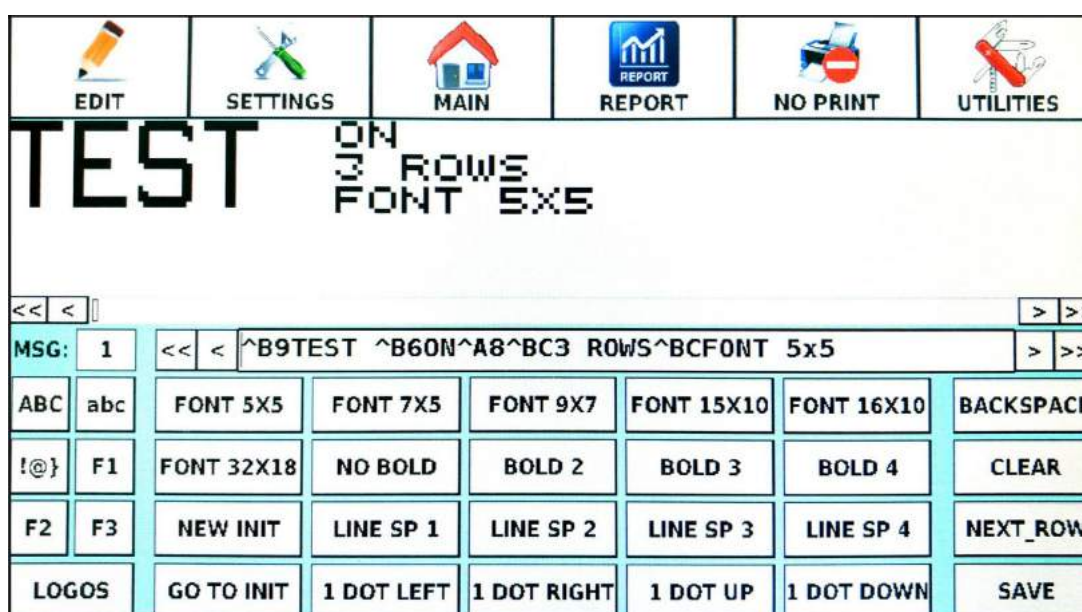
Using the buttons and moving between the fields is very easy and does not necessitate any complex explanations.

Each subject will be demonstrated by a pictutre example which will provide sufficient information.

What You See Is What You Get (WYSIWYG): the display will show an exact replica of what will be printed.

The following appears on the screen upon start-up:

4.1 INITIAL SCREEN



The image of the hypertext being printed is displayed along with a bar containing its sequential numerical code (001), the content of the hypertext (^B9TEST), its report (number of prints).

If manual printing is selected, the Printing image may be altered and another one may be selected by touching that bar.

If automatic printing (BCD) is selected, no further modifications are possible and the selection of the hypertext to be printed is done automatically by the BCD inputs.

It is possible to move from screen to screen by touching the icon buttons.



Allows entering into the first screen of the editor.



Enters the first screen of the setting menu.



Allows returning to the initial screen from any other screen.



Enters the report display page.



Activates / deactivates the print operation.

Shows a new menu:



Which allows other functions.



Activates a purge operation on the print head.



Shows the monitor page (same page that can be opened by touching UTILITIES and MONITOR).



Shows that the smart card is not inserted and / or the ink credit is exhausted.

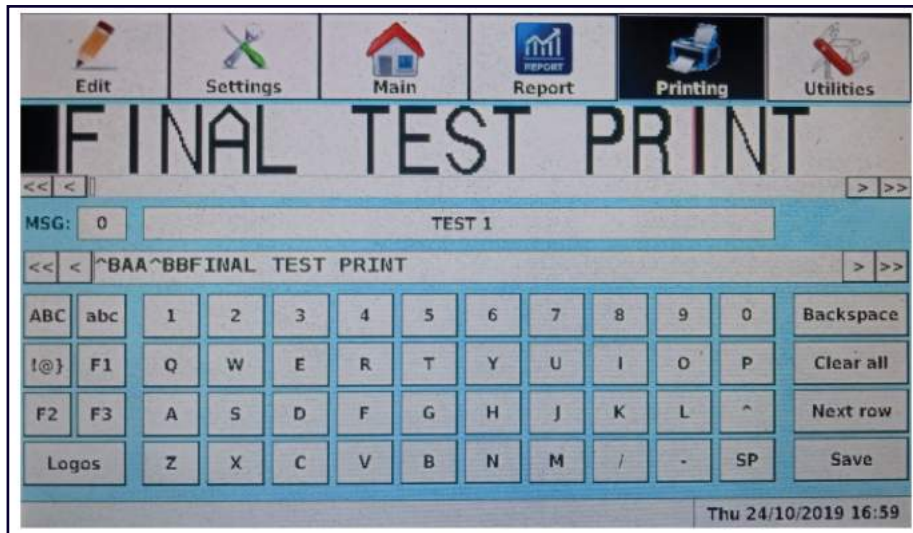


Shows the number of the remaining prints for the selected hypertext shown, when the smart card is inserted,

The bottom bar shows warnings, alarms and information for the operator.

On the left side of the bar, date and time are shown. By touching here, it is possible to change them.

4.2 SCREEN: EDIT



- In the EDIT page it is possible to create a combination of texts, barcodes, logos, time / date codes, expirations dates, counters, shifts.
- Some of these are fixed, some are programmable to automatically generate variable texts.
- This combination generates a dot image (BITMAP) that represents the image that will be printed and which is shown on the display.
- Such combination will be called henceforth HYPERTEXT and the system can manage and store 200 of such texts.
- The largest bitmap that the system can manage for printing is 5000 columns.
- This size cannot be exceeded. If the stored bitmap is larger, the printing will be truncated.
- An **INSERTION BAR** is shown below the bitmap and any time a key is pressed a character or a group of characters is added or inserted in the bar.
- The maximum number of characters that can be inserted into the insertion bar is 256.
- All the available functions of the hypertext are realized by inserting Special Characters.
- They are shown on the insertion bar as groups of 2 characters in hexadecimal format preceded by the character ^.

For example ^B9 indicates that the 9x7 font has been selected, ^C4 will print the number of the day.

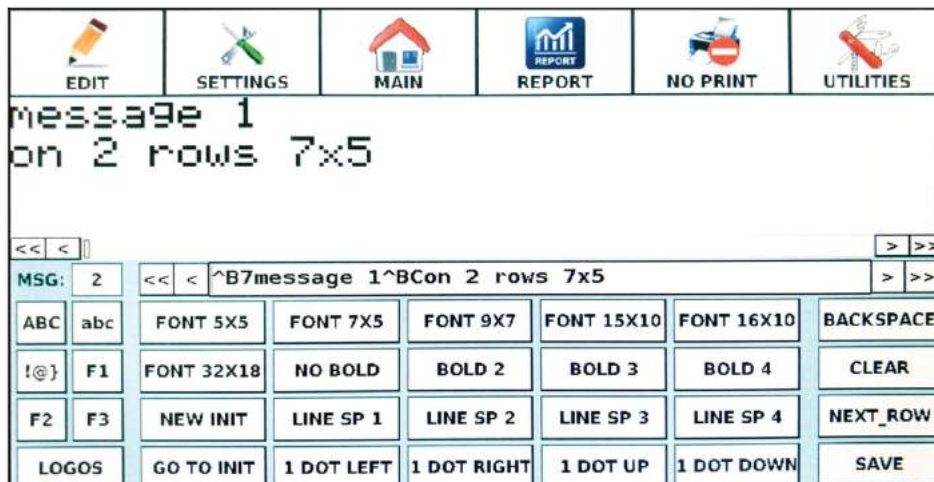
- The list of these Special Characters is given in the next pages.
- By touching the button next to **MSG:** the number of the hypertext is selected.

EXAMPLE

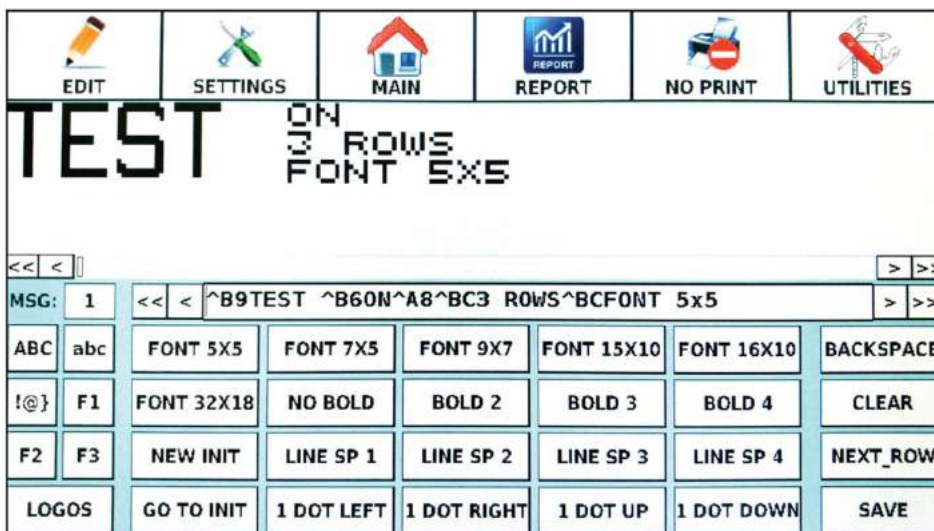
Insertion of two texts with two different fonts.

First press **CLEAR** to delete any previous image.

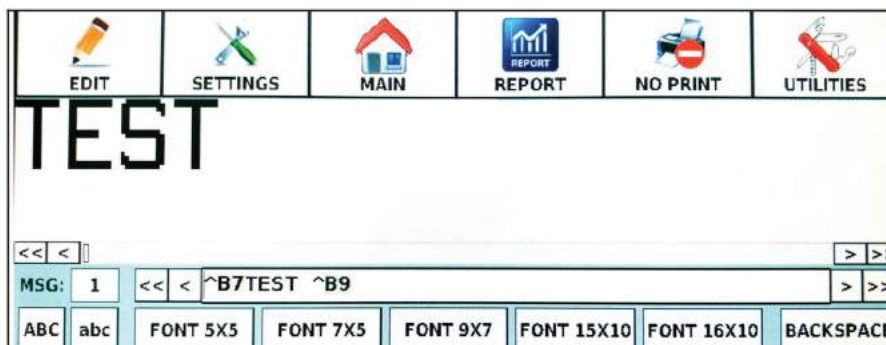
Then press **FONT 7x5**: the Special character ^B7 will be added.



Then insert the text "TEST":



Then press **FONT 9x7**: the Special character ^B9 will be added.



Then insert the text “ PRINT ”:

 EDIT	 SETTINGS	 MAIN	 REPORT	 NO PRINT	 UTILITIES
TEST PRINT					
<< < > >>					
MSG: 1	<< < ^B7TEST ^B9PRINT > >>				
ABC abc	FONT 5X5	FONT 7X5	FONT 9X7	FONT 15X10	FONT 16X10 BACKSPACE

By pressing **SAVE**, the hypertext no.12 is ready to be printed as shown on the display.

The various buttons on the left allow to create texts, logos, dates, time, expirations dates, counters, shifts, as described in the next pages.

“ABC” letters in upper case

MSG: 7	<< < ASDFGHJK123456 > >>											
ABC abc	1	2	3	4	5	6	7	8	9	0	BACKSPACE	
!@}	F1	Q	W	E	R	T	Y	U	I	O	P	CLEAR
F2	F3	A	S	D	F	G	H	J	K	L	^	NEXT_ROW
LOGOS		Z	X	C	V	B	N	M	/	-	SP	SAVE

“abc” letters in lower case

MSG: 7	<< < ASDFGHJK123456 > >>											
ABC abc	1	2	3	4	5	6	7	8	9	0	BACKSPACE	
!@}	F1	q	w	e	r	t	y	u	i	o	p	CLEAR
F2	F3	a	s	d	f	g	h	j	k	l	^	NEXT_ROW
LOGOS		z	x	c	v	b	n	m	/	-	SP	SAVE

Note : to print the character ^, press ^ twice.

!@} non-letter characters

MSG: 7	<< < ASDFGHJK123456 > >>											
ABC abc	1	2	3	4	5	6	7	8	9	0	BACKSPACE	
!@}	F1	!	"	#	\$	%	&	'	(	)	*	CLEAR
F2	F3	+	,	.	:	;	<	>	=	?	@	NEXT_ROW
LOGOS		[	]	\	_	{	}		~	`	SP	SAVE

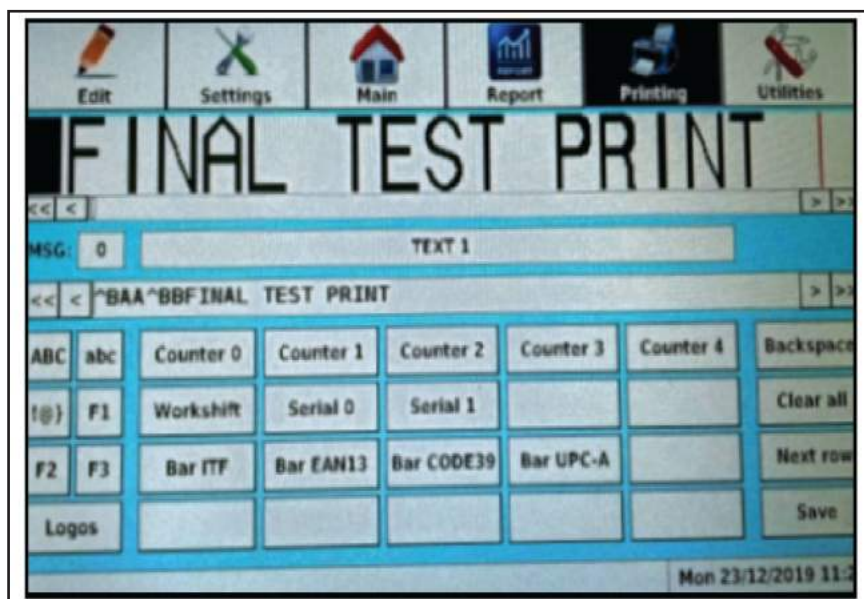
F1 selection of fonts, bold, interline spacing, text movement

MSG:	9	<<	<	^B65*5^B77*5^B89*7^D315*10^B916*10^D432*18												>	>>
ABC	abc	FONT 5X5		FONT 7X5		FONT 9X7		FONT 15X10		FONT 16X10		BACKSPACE					
!@}	F1	FONT 32X18		NO BOLD		BOLD 2		BOLD 3		BOLD 4		CLEAR					
F2	F3	NEW INIT		LINE SP 1		LINE SP 2		LINE SP 3		LINE SP 4		NEXT_ROW					
LOGOS		GO TO INIT		1 DOT LEFT		1 DOT RIGHT		1 DOT UP		1 DOT DOWN		SAVE					

A printable area 32-dot high is provided.

- In this area up to 5 lines of text can be inserted and 6 different types of font matrix are available : 5x5, 7x5, 9x7, 15x10, 16x10 and 32x18.
- Considering the height of the printable area, more or less lines of text can be printed depending on the font selected.
- **For example:**
 - up to 5 lines of text with 5x5 font and line spacing 1
 - up to 4 lines of text with 7x5 font and line spacing 1
 - up to 2 lines of text with 15x10 font
 - T1 line of text with 32x18 font
- To move to the next line of text, press the button NEXT ROW. The selected interline spacing will be used.
- It is possible to “ multiply ” the columns being printed to obtain print that is larger (BOLD 2, BOLD 3, ...).
- It is possible to move the text by one dot in the 4 directions (1 DOT LEFT, 1 DOT RIGHT, ..).
- It is possible to change the interline spacing between two lines of text (LINE SP1, LINE SP 2, ...).
- The text which is added is positioned immediately after the previous one, but it is possible to move it to the initial column (GO TO INIT) or to move the initial column to the present position (NEW INIT).

F2 insertion of counters, work shift, barcodes, serial port



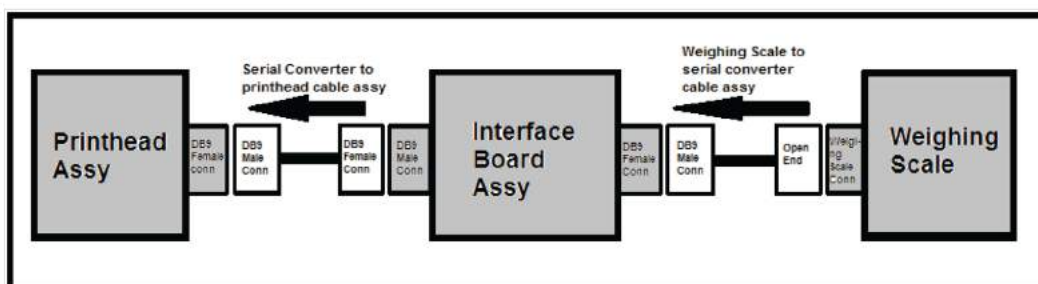
Five product counters are available, either displaying letters or numbers (increasing or decreasing) which also serve batch counting purposes.

They can be inserted into the hypertext (COUNTER 0, COUNTER 1, ..).

Reading inputs from weighing scale feature is also available.

Usage: For applications related to printing of weight values at customer end, by taking inputs directly from the weighing scale in real time basis. Serial 0 and serial 1 commands are to be used for reading of the inputs retrieved from the weighing scale.

Connection Block Diagram:



DB 9 Male Pin No.	Pin connections	Weighing Scale Point
Pin no. 2	R _x	T _x point of above to R _x
Pin no. 3	T _x	R _x point of above to T _x
Pin no. 5 & 9 (shorting)	GND	GND point

Above table shows the connections of the weighing scale to serial converter cable assy.

Data format for the above setup is as <Data><Carriage Return><Line Feed>.

Data queuing: It means, that the printer will keep storing i.e. will be queuing the information received on FIFO basis from the weighing scale,

For example, if two consecutive weight inputs are received by the printer of two different products passing the weighing scale, then the printer will start queuing the information and wait till the products to pass through the printhead to print on it.

Information will get printed purely on “FIFO” basis.

DIP Switch setting in interface board assy:

Switch 1 and 4 will be used and factory setup level, at customer places both shall remain “OFF”

Switch 2 is “weighing scale mode” and it should be “ON” always.

Switch 3 is for “Echo mode”. and it could be “ON” or “OFF” depending upon requirement”

When turned “ON” String received by printer will be reflected back as an acknowledgement.

Customer can also save this data so as to keep record of weights/strings printed on daily basis. When turned “OFF” printer will not send any feedback/acknowledgement.

N.B.: All the above features and functions, mentioned above might not be functional with software releases below 2.00.

Their dedicated purpose and utility needs to be formatted in the SETTINGS->COUNTERS page (see page 42 below).

A shift number, with up to 4 daily intervals, can be associated with an alphanumeric code and inserted into the hypertext by pressing.

WORKSHIFT: Its dedicated purpose and utility needs to be formatted in the SETTINGS->WORKSHIFT page (see page 41 below).

Four different barcodes can be inserted into the hypertext:

- ITF, Interleaved Two of Five
- EAN 13
- Code 39
- UPC-A

The specifications for these barcodes have to be followed while typing their content.

For example, the EAN13 barcode is only numerical, thus letters cannot be inserted.

F3 insertion of dates, time, expirations

ABC	abc	NUM MONTH	ALPHA MONTH	YEAR 1 DIG	YEAR 2 DIG	YEAR 4 DIG	BACKSPACE
!@}	F1	DAY	MONTH	YEAR	JULIAN DAY	WEEK DAY	CLEAR
F2	F3	WEEK	SECONDS	MINUTES	HOURS		NEXT_ROW
LOGOS		EXP DAY	EXP MONTH	EXP YEAR			SAVE

Day, month, year, week, sequential day, expiration, hours, minutes, seconds in any format can be inserted into the hypertext by touching the corresponding button.

Expiration has to be previously formatted in the SETTINGS->EXPIRATION page.

NUM MONTH and **ALPHA MONTH** determine the way the month will be displayed / printed when pressing the **BUTTON MONTH**.

Months can be printed as a number (1-12) or in letters (JAN-DEC).

YEAR 1 DIG, **YEAR 2 DIG**, **YEAR 4 DIG** determine the way the year will be printed when pressing the button YEAR. Years can be printed in 1, 2 or 4 digit.

LOGOS: insertion of logos

MSG: 7		<< <		ASDFGHJK123456										> >>	
ABC abc		1	2	3	4	5	6	7	8	9	0	BACKSPACE			
!@} F1			→	←	↑	↓	◊	▲	☪	↕	⌈	CLEAR			
F2 F3		—	]	ä	ö	ü	€	ñ	ä	ö	^	NEXT_ROW			
LOGOS		ü	¢	ñ	ä	ö	ü	¢	/	.	SP	SAVE			

Logos shown on the keyboard have to be previously created in the UTILITIES->LOGOS page.

Logos are printed depending on the last font format selected.

4.2.1 SPECIAL CHARACTERS

Special characters are needed to access certain functions i.e. expiration, date, time, counters, shift, logo, modify the font type and move the objects within in the text being printed.

They are shown as groups of 2 characters in hexadecimal format preceded by the character ^.

They are automatically generated by pressing the related key (for example, by pressing the **FONT 9X7** key, the special character ^B9 is inserted), but they can also be inserted by typing the 3 characters manually.

^9F Go to bitmap in it:	Moves the input point of the next character to the bitmap origin. Can be used to overwrite a logo with a text first insert the logo and then, after the ^9F special character, enter normal text. The effect will be to write over the logo.
^A0 Not used:	
^A1 Not used:	
^A2 Not used:	
^A3 Set new bitmap in it:	Moves the start of the bitmap to the current position. Can be used many times and is useful when working with several lines of text.
^A4 Move one dot forward:	Moves the current dot forward by one column. Can be used more than once and allows high precision printing, e.g. when using texts and logos.
^A5	As above, but moves the dot back by one column.
^A6	As above, but moves the dot up by one row of dots.
^A7 Move one dot up:	As above, but moves the dot down by one row of dots.
^A8 Line Spacing 1:	Sets the space between two text lines by 1 dot. Can be used when 5 lines of 5x5 font need to be printed.
^A91 Line Spacing 2:	Sets a line spacing of 2 dots It is the default setting.
^AA Line Spacing 3:	As above, line spacing set to 3.
^AB Line Spacing 4:	As above, line spacing set to 4.
^AC Line Spacing 5:	As above, line spacing set to 5.
^AD No bold:	Standard text (default).

^AE Bold 2:	The columns of the printing are doubled.
^AF Bold 3:	The columns of the printing are multiplied by 3.
^B0 Bold 4:	The columns of the printing are multiplied by 4.
^B1 Months in digits:	Formats the system to print the month in digits (112).
^B2 Months in letters:	Formats the system to print the month in letters (JAN, FEB, ..) depending on the language currently selected.
^B3 1-digit year:	Formats the system to print the year with one digit (0-9).
^B4 2-digit year:	Formats the system to print the year with two digits (00-99).
^B5 4-digit year:	Formats the system to print the year with four digits.
^B6 5x5 font:	Formats the use of the 5x5 font. Up to 5 lines of text.
^B7 7x5 font:	Formats the use of the 7x5 font (default). Up to 4 lines of text.
^B8 9x7 font:	Formats the use of the 9x5 font. Up to 3 lines of text.
^B9 16x10 font:	Formats the use of the 16x10 font. One single line of text.
^BA Start logos:	After this special character, each capital letter will represent a logo.
^BB End of logos:	This character ends the logo sequence. The pair start-end can be used more than once in a text message.
^BC Move to next line:	Performs the “return” function, the text moves to the next line.
^BD Barcode ITF init:	Only an even number of digits are allowed. To terminate insert a space.
^BE Print counter nr.0:	Enables the automatic printing of the first counter. It will be printed in the way formatted in the counter page.
^BF Print counter nr.1:	As above: second counter.
^C0 Print counter nr.2:	As above: third counter.
^C1 Print counter nr.3:	As above: fourth counter.

^C2 Print counter nr.4:	As above: fifth counter.
^C3 Not used:	
^C4 Print day:	Automatic printing of the day (1-31).
^C5 Print month:	Automatic printing of the month in letter or digits depending on the way it was formatted using the special characters ^B1 and ^B2.
^C6 Print year:	Automatic printing of the year with 1, 2, 3 or 4-digit depending on the way it was formatted using the special characters ^B3, ^B4 and ^B5.
^C7 Print fiscal week:	Automatic printing of the week (1-53)
^C8 Print week day:	Automatic printing of the week day (0-6, with 0=Monday).
^C9 Print Julian day:	Automatic printing of Julian date (1-365 or 1-366 on leap years).
^CA Print seconds:	Automatic printing of seconds (1-59).
^CB Print minutes:	Automatic printing of minutes (1-59).
^CC Print hours:	Automatic printing of hours (0-23).
^CD Expiration day:	Automatic printing of day, month and year of expiration as set in the relevant menu page.
^CE Expiration month:	Automatic printing of month and year of expiration as set in the relevant menu page.
^CF Expiration year:	Automatic printing of expiration year as set in the relevant menu page.
^D0 Print shift code:	Automatic printing of current shift code as set in the relevant menu page.
^D1 Not used:	
^D2 not used:	
^D3 15x10 font:	Selects the use of the 15x10 font. Up to 2 lines of text.
^D4 32x18 font:	Selects the use of the 32x18 font. One single line of text.

- ^D5 Barcode EAN13 in it: Only 13 digits are allowed. To terminate insert a space. Check digit is automatically calculated.
- ^D6 Barcode CODE 39 in it: Digits and letters are allowed. To terminate insert a space.
- ^D7 Barcode UPC-A in it: Only 12 digits are allowed. To terminate insert a space. Check digit is automatically calculated.

4.3 SCREEN: SETTINGS

After pressing SETTINGS, the first screen related to the PRINT is shown:

• Right column 300dpi

Only the 150 nozzles of the right column will be activated producing 300dpi print.

• Left column 300dpi

Only the 150 nozzles of the left column will be activated producing 300dpi print.

• Both columns 600dpi

Both the 150 nozzles of the right and left column will be activated producing 600 dpi print.

• Column number

Starting from the first column (on the left) define how many columns of the image are to be printed.

Use values from 1 to 2000.

A vertical red line will indicate the part of the image that will be printed.

Everything on the right hand side of the red line (text, barcode, expiry, time, turn, etc.) will not be printed.

- **Inter column**

To be modified based upon the “passing” speed of the product to be printed.

The faster the product moves the lower this value must be.

The value to be inserted is determined by practical printing tests depending on the type of object to be printed on, the passing speed of the product and the desirable print results. Values from 0 to 65535, each value is equal to 1µs (microsecond). When an encoder is used, each value corresponds to an encoder pulse.

- **Print delay**

Once the photocell is triggered by the passing of the product to be printed has been positioned, this parameter is used to delay the start of the printing, and, consequently, The correct position of the printed image on the product.

Values from 1 to 65535, each value is equal to 1ms (millisecond).

When an encoder is used, each value corresponds to an encoder pulse.

- **Bold**

By increasing this number it is possible to replicate the printing of each printed column of dots.

Images appear darker by increasing this number and by decreasing the Inter column spacing.

Values from 1 to 9.

- **Column offset**

This value is used to synchronize print outputs from both printing columns to get a sharp image.

- **Print repeater**

Enable / disable the print repeater.

Once the start printing photocell has been engaged printing will be repeated whilst the photocell remains active.

The first print follows the timing set in the Print delay field. The next prints will occur at intervals based on the time set in the Repeat time field.

- **Repeater time**

In the case of continuous or repeated printing, this defines the time between the first and the subsequent prints values from 1 to 65535, each value is equal to 1ms. When an encoder is used, each value corresponds to an encoder pulse.

- **Left to right**

Determines the print direction of the product to be printed.

- **Inverse print**

The image will be printed upside-down.

- **Border lines**

Allows the printing of a top and a bottom line.

- **Encoder**

Enable / disable the use of the encoder.

Once the encoder has been enabled the parameter Inter column becomes the number of encoder pulses between one column and the next.

Start at a low value (5-10) and then increase it to enlarge the printed image.

- **Clockwise rotation**

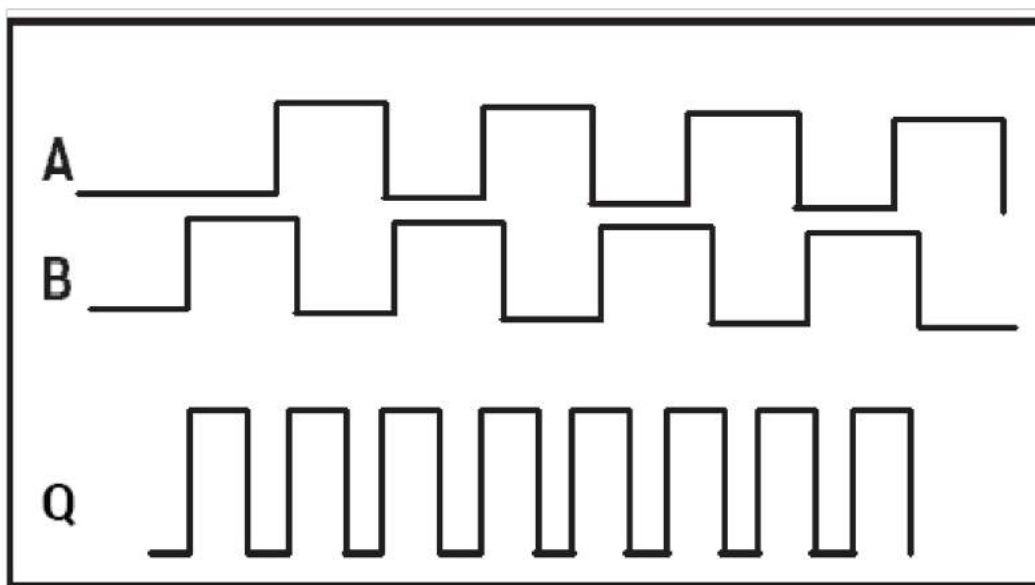
Formats the rotation direction of the encoder.

Enable this if the encoder rotates clockwise.

When Encoder is enabled, a value is shown in the MONITOR page which allows to check if the encoder signal is properly received (value changes when rotating encoder). This value should increase. If it decreases, change this setting.

- **Encoder by 4**

By enabling this, the two signals (A and B) of the encoder are combined to multiply by 4 the number of the pulses. See Q signal in illustration.



- **Zero without slash**

Formats the printing of zero with or without slash.

- **Product detection**

Internal: when the built-in photocell is to be used

External: when an external photocell is connected to the printhead.

Inverted: when the photocell operates as “normally activated”

On level: when the print should immediately stop when the photocell is deactivated

- **Macrodot size**

This parameter allows the height of the printed image to be reduced and at the same time the height of each font from their standard sizes:

- Font 5x5 2.4mm
- Font 7x5 3.0mm
- Font 9x7 3.6mm
- Font 15x10 6.0mm
- Font 16x10 6.4mm
- Font 32x18 12.7mm

For example, FONT 32x18 can range from 12.7 to 1.5 mm:

Macrodot size 0: Normal height 12.7mm

Macrodot size 1: Character height 1.5mm

Macrodot size 2: Character height 2.8mm

Macrodot size 3: Character height 4.0mm

Macrodot size 4: Character height 5.3mm

Macrodot size 5: Character height 5.6mm

Macrodot size 6: Character height 6.9mm

Macrodot size 7: Character height 8.5mm

Macrodot size 8: Character height 9.2mm

Macrodot size 9: Character height 10.8mm



Note: Aspect ratio of the characters can be adjusted by changing inter column setting.

- **BCD Enable**

This setting enables automatic selection of messages through BCD signal selection.

- **BCD Strobe**

When this setting is enabled, printer will only check for BCD selection, if the signal is high. For example: If strobe signal is high, and BCD 0 and BCD 1 are also at high at same time, then message selected will be 003.

If BCD strobe is not high, and BCD 0 and BCD 1 are high, then printer will not check for any message selection. It will continue with the previously selected message.

- **BCD Texts**

BCD texts indicates the printer, how many messages are there, which are available for BCD selection. Keeping this value low makes BCD recodification faster.

For example : If there are 03 messages to print, it is recommended to put 03 in BCD text window.

- **BCD Block**

Block refers to set of 16 messages.

For example: set of messages 1-16 are block 0. Set of messages 17-32 are block 02 and so on. BCD block "0" means that we are considering messages 1-16 for printing through BCD selection.

• 150 dpi

TIJ printer is capable to print at lower resolutions. Enabling 150 dpi means, printer will print from every alternate nozzle, delivering 150 dpi print vertically.

This option is an important tool where customer wants to reduce per print cost by compromising with the print quality.

When any change is done in this page or the next pages, press **SAVE** to store the new setting into the printhead.

After pressing **QUALITY** in the **SETTINGS** page the following screen will be displayed:

• Auto Quality:

When enabled the printhead automatically assumes the default settings for quality parameters (print voltage, pulse width) from the smart card that is used with the printhead. Hence, eliminating the need for manually setting the quality parameters.

The screenshot shows a printer's settings interface with a teal background. At the top, there is a navigation bar with icons and labels: Edit (pencil), Settings (wrench), Main (house), Report (bar chart), No print (printer with red X), and Utilities (wrench and screwdriver). Below this, on the left, is a vertical menu with buttons: Print, Quality (highlighted), Output, Date & time, Expiration, Workshifts, Counters, Reset, and Terminal. The main area on the right contains the following settings:

- ☐ Auto quality
- 116 Print voltage
- 2100 Pulse width
- ☐ Warming
- 0 Temperature setpoint
- ☐ Autospit
- 0 Autospit minutes
- 0 Autospit columns
- ☐ Pre-heating waveform
- ☐ Voltage boost

A 'Save' button is located at the bottom right of the settings area. At the bottom of the screen, a status bar shows the date and time: 'Wed 13/01/2020 15:55'.

• Print voltage :

Represents the value of the voltage used to drive the print head. It determines to the quantity of ink fired through the nozzles.

The value is shown in dV (tens of Volt), for instance, 120 means 12.0 V. Higher values make the print darker.

Always use the voltage which is automatically set by the smart card which corresponds to optimum performance or set firing parameters manually as suggested on cartridge label.

****always use specified firing parameters, usage of higher voltages and Pulse width may lead to cartridge resisters cogation and thus bad print quality/damage of cartridge resisters.**

- **Pulse width:**

Represents the width of the pulse used to drive the print head. Cartridge type and ink viscosity determine this value.

The value is shown in ns (nano seconds), for instance, 1840 means 1840 ns corresponding to 1.84µs.

Always use the driving pulse width which is automatically set by the smart card or the value specified on cartridge label which corresponds to ideal printing conditions.

- **Warming:**

When enabled, the nozzles which are not “used” for printing are driven with a 400 ns pulse which maintains the ink warm.

Temperature set point operating temperature in °C of the ink.

- **Autospit:**

When enabled, this function prints one or more full columns of dots at set intervals.

Autospit minute's period, is minutes between full column print.

Autospit columns number of full column fired during auto spit.

- **Pre-heating waveform:**

When enabled, the standard driving pulse is preceded by an “extra” pulse (About 400ns long).

This feature tends to eliminate the side effects of short decap times in certain inks but could decrease the maximum print speed.

**recommended to be used only when ambient temperature is near to 0°C

- **Voltage boost:**

When enabled, the first print after at least a 5 minute “no-print” period is driven with the maximum voltage (about 14V).

After pressing **DATE & TIME** in the SETTINGS page another screen is shown in which it is possible to set time and date of the printhead and of the terminal and synchronize them. By inserting in a hypertext the required special character, automated printing of date and time in any format can be obtained.

EDIT	SETTINGS	MAIN	REPORT	NO PRINT	UTILITIES
PRINT	Date on terminal: 10 / 04 / 2018 Time on terminal: 12 : 40 : 02 SAVE				
QUALITY					
DATE & TIME					
EXPIRATION					
WORKSHIFTS					
COUNTERS	Date on printhead: 10 / 04 / 2018 Time on printhead: 12 : 23 : 06				
RESET	Date rollover time: 0 : 0 SYNC SAVE				
TERM					

EXPIRATION

- After pressing EXPIRATION in the SETTINGS page another screen is shown in which the expiration date can be formatted.
- It is possible to define expiration dates and insert them into hypertexts. They can be counted in years, months or days.
- If the expiration is in years (0-9) the system will print the year only.
- E.g., with a 2 years expiration and the related special character (^CF), the system will print the current year increased by two.
- The expiration year will be printed in the selected format (1-, 2- or 4-digit) depending on the special character used into the hypertext.

EDIT	SETTINGS	MAIN	REPORT	NO PRINT	UTILITIES
PRINT	365 Days	Format: ☐ MM YY DD - MM YY ☐ YY MM DD - YY MM ☐ MM DD YY ☐ YY DD MM ☒ DD MM YY ☐ DD YY MM			
QUALITY	12 Months				
DATE & TIME	1 Years				
EXPIRATION	Separator: ☐ / ☐ . ☒ Space				
WORKSHIFTS					
COUNTERS					
RESET					
TERM	SAVE				

- If the expiration is in months (00-99), the system will print year and month or month and year depending on the selection (YY-MM or MM-YY).
- E.g. with a 6 months expiration and the special character ^ CE the system will calculate the month and year of expiration and print these in the selected format (month / year or year / month).
- The year is printed in 1- 2- or 4-digit format depending upon the special character inserted into the hypertext.
- The month will be printed in digits or letters depending upon the relevant special character and the selected language.
- It is also possible to select the character to be used as separator (/ . Space).
- The various options allow the operator to print the expiration date of e.g. May 2006 as:
MAY 2006 06.MAY 5/6 MAY/6 MAY/06 MAY 2006
2006/5 etc.
- With an expiration in days (000-999), the system will print year, month and day.
- The format can be selected as follows:

month	year	day
year	month	day
month	day	year
year	day	month
day	month	year
day	year	month

- E.g., with a 200 days expiration and the special character ^CD, the system will calculate the day, month and year of expiration adding 200 days to the current day, and print the resulting date in the selected format.
- The year is printed in 1- 2- or 4-digit format depending upon the special character inserted into the hypertext.
- The month will be printed in digits or letters depending upon the relevant special character and the selected language.

- It is also possible to select the character to be used as separator.
- Some of the ways the system can print the expiration date 10 May 2006, on the basis of all the various possibilities are:
 - 10 MAY 2006 10.06.MAY 5/6/10 10/MAY/6 MAY/10/06 10 MAY 2006 10.2006.5 etc.
- The settings available at this page will be used on the basis of the special character inserted into the hypertext.
- e.g: By inserting the special character ^CD (expiration days), the system will take into account the set number of Days (45 in the picture) and will not use the information relevant to Months (2) and Years.

WORKSHIFTS

- After pressing WORKSHIFTS in the SETTINGS page another screen will appear in which the work shifts can be formatted.
- Up to 4 shifts, identifiable by the start hour and a code which will be printed during the shift itself, may be introduced.
- There are 4 shifts in the example given on the side from 8 to 12, from 12 to 18, from 18 to 22, from 22 to 8 on the following day. The codes A, B, C, D, are related to these shifts.
- Practically from 8 to 12, the code A will be printed from 12 to 18, the code B will be printed etc.

 EDIT	 SETTINGS	 MAIN	 REPORT	 NO PRINT	 UTILITIES										
PRINT	<table border="1"> <thead> <tr> <th>START TIME</th> <th>CODE</th> </tr> </thead> <tbody> <tr> <td>08 : 00</td> <td>A</td> </tr> <tr> <td>12 : 00</td> <td>B</td> </tr> <tr> <td>18 : 00</td> <td>C</td> </tr> <tr> <td>22 : 00</td> <td>D</td> </tr> </tbody> </table>					START TIME	CODE	08 : 00	A	12 : 00	B	18 : 00	C	22 : 00	D
START TIME	CODE														
08 : 00	A														
12 : 00	B														
18 : 00	C														
22 : 00	D														
QUALITY															
DATE & TIME															
EXPIRATION															
WORKSHIFTS															
COUNTERS															
RESET															
TERM															
SAVE															

- There will be no more shifts when the start hour of one shift is identical to that preceding it.

- For example, making the start hour of shift 3 the same as that of shift 2 will result in printing only the first two shifts.

COUNTERS

- After pressing COUNTERS in the SETTINGS page another screen will appear in which up to 5 counters can be formatted.

Counters can be inserted into the hypertext and then printed.

 EDIT	 SETTINGS	 MAIN	 REPORT	 NO PRINT	 UTILITIES
PRINT	0	Selected Counter			
QUALITY	☐	Batch			
DATE & TIME	55	Current Value			
EXPIRATION	999999999	Max Value			
WORKSHIFTS	1	Step			
COUNTERS	☐	Decremental			
RESET	☐	With zeroes			
TERM	☒	Alphabetic			
	9	Digits to print			
	0	Quantity now			
	1	Quantity set			
SAVE					

- **Selected Counter:**

0-4, to select the counter that requires formatting

- **Batch (lot):**

If selected, this counter is used to “calculate” the number of items inside a batch. The lot number is set with Current Value, its quantity with Quantity Value.

- **Current Value:**

0-999999999, the value of the counter at present (e.g. the number of products already printed).

- **Max Value:**

0-999999999, the limit value of the counter. If the counter is increasing, it will reset to zero once it reaches the Max Value. If the counter is decreasing, the counter will reset to this value after reaching zero.

- **Step:**

0-255, the “size” of the step the system makes each time a print is made.

- **Decremental:**

Decrementing counter (counting down) when selected, otherwise the counter increments (counting up).

- **With zeroes:**

The system prints also the zeros on the left but are not significant when selected.

- **Alphabetic:**

Prints digits (0-9) or prints letters (A-Z) if selected. In this case A is equivalent to 0 (zero).

- **Digits to print:**

The number of digits of the counter that will be printed (up to 9).

- **Batch counter:**

When selecting a Batch counter you will see:

 EDIT	 SETTINGS	 MAIN	 REPORT	 NO PRINT	 UTILITIES
PRINT	0	Selected Counter			
QUALITY	☐	Batch			
DATE & TIME	55	Current Value			
EXPIRATION	999999999	Max Value			
WORKSHIFTS	1	Step			
COUNTERS	☐	Decremental			
RESET	☐	With zeroes			
TERM	☒	Alphabetic			
	9	Digits to print			
	0	Quantity now			
	1	Quantity set			
SAVE					

- **Quantity now:**

Is the value reached inside a batch.

- **Quantity set:**

Is the value set as batch quantity.

The batch counter prints the same code for a set number of times (batch quantity) then increases it.

The value of each counter is stored in the system memory every time the counter changes.

RESET

After pressing RESET in the SETTINGS page a confirmation will be required to erase the entire memory of the printer.

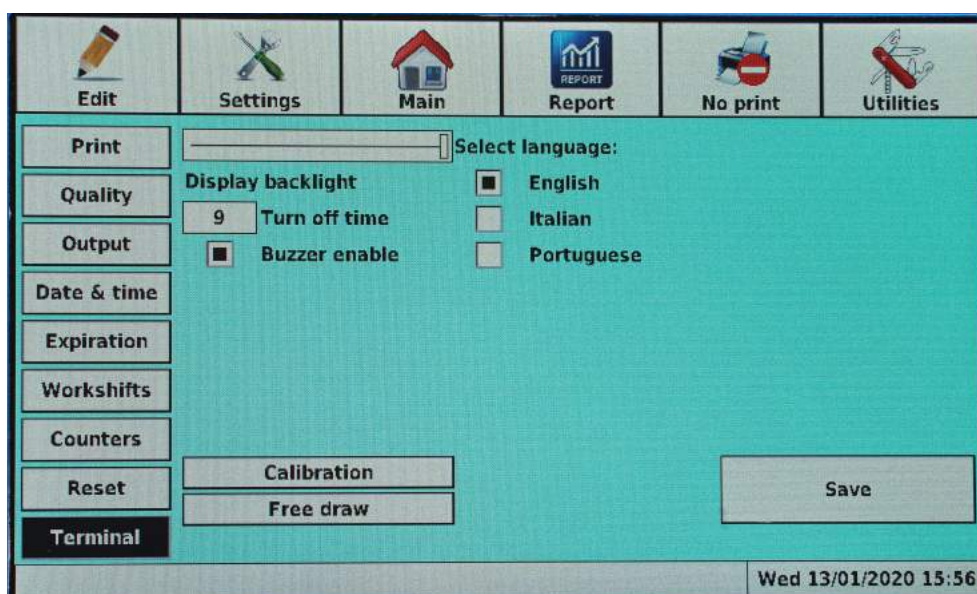
Only press the button shown below if really it is necessary to erase everything.

PRESS TO RESET OF PRINTHEADS

ATTENTION: This operation should be performed carefully. It would not be possible to recover the data which has been erased.

TERM

After pressing TERM in the SETTINGS page another screen will appear in which it is possible to format:



- **Display backlight:**

Reduce in dark, increase in light environments.

- **Display turn off time:**

To save energy and to avoid unnecessary heating of the terminal, the backlight is switched off after a set time.

*recommended to use lower most value as per user convenience.

- **Buzzer enable:**

A beep can be enabled when touching the screen.

From this page it is also possible to run the calibration of the touch screen and to test it using a free draw tool.

- **Select Language:**

Select the language of the terminal from the options given.

- **Calibration:**

For proper touch and functioning of the terminal, calibration is required. It is a four point calibration. Touch the points properly using the stylus, for proper calibration of touch calibration.

- **Free Draw:**

Free draw allows us to cross check the calibration and determine whether calibration done was fruitful or not.

4.4 SCREEN: REPORT

From this screen it is possible to analyze how many times any individual hypertext has been printed and therefore “Products manufactured”.

It is also possible to zero-out the reports either together or individually.

In order to zero-out any single report, select it and press the **CLEAR SELECTED REPORT** button.

To zero-out all the reports, press the **CLEAR ALL** button.

Reports are “saved” after each print to avoid loss of data in case of power-off.

 EDIT	 SETTINGS	 MAIN	 REPORT	 NO PRINT	 UTILITIES																						
<table border="1"> <thead> <tr> <th>000</th> <th>000000</th> </tr> </thead> <tbody> <tr><td>001</td><td>000000</td></tr> <tr><td>002</td><td>000000</td></tr> <tr><td>003</td><td>000000</td></tr> <tr><td>004</td><td>000000</td></tr> <tr><td>005</td><td>000000</td></tr> <tr><td>006</td><td>000000</td></tr> <tr><td>007</td><td>000000</td></tr> <tr><td>008</td><td>000000</td></tr> <tr><td>009</td><td>000000</td></tr> <tr><td>010</td><td>000000</td></tr> </tbody> </table>						000	000000	001	000000	002	000000	003	000000	004	000000	005	000000	006	000000	007	000000	008	000000	009	000000	010	000000
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010	000000																										
CLEAR SELECTED REPORT CLEAR ALL																											

4.5 SCREEN: MONITOR

From this screen it is possible to analyze the status of the inputs, the ink credit of the smart card, the software releases, and working hours.

This information should be analyzed in case of maintenance or service.

 Edit	 Settings	 Main	 Report	 Printing	 Utilities
0	Encoder	0F	Head state	31634	Msg dots
☐	Ext start	05	Prn state	31560	Printed dots
☐	Pogo start	0048	CTRL PRN	999820412	Credit
☐	FPGA printing	00	RAM STAT	999999999	Full credit
☐	BCD strobe	00	STAT PRN	38	Working hours
0	BCD value	☒	RAM ok		
995	TSR	☒	RTC ok		
26	Head temp.	7" terminal rel. 2.04			
27	Pogo temp.	TIJmini printhead			
		SN: 000000			
		uC rel. 2.04			
35	Term temp.	FPGA rel. 2.02			
Sat 26/10/2019 11:22					

Encoder: In monitor image box in front of encoder represents, encoder pulses count. If this value is changing, that means encoder is working and pulses are being sent to printer properly.

However this value should increase always, if not so, please toggle encoder direction.

External Start: When any trigger from external photocell is received, black dot appears in front of this field. This information can be used to check if external photocell is working ok or not?

POGO Start: When any trigger from internal photocell is received, black dot appears in front of this field. This information can be used to check if internal photocell is working ok or not?

FPGA PRINTING: Black dot appear in front of this field when Print is in progress.this information can be used if print output is not received.
This will indicate whether parameters set are not proper/cartridge is faulty or print head electronics is faulty.

BCD STROBE: Black dot appears in front of this field when a strobe signal is received.

BCD VALUE: This field represents current message number selected by BCD signals.

TSR: This represents TSR value for cartridge.

HEAD TEMP: This represents current temperature of printhead/cartridge.

POGO TEMP: This represents ambient temperature/POGO board electronics temperature.

TERM TEMP: This represents current temperature for 7" terminal electronics.

****Other status like HEAD STATE, PRN STATE, CTRL PRN, RAM STAT, STAT PRN, RAM OK, RTC OK, are for official debugging.**

MSG DOTS: This value represents number of pixels used in current messages being printed.

PRINTED DOTS: This value represents number of pixels/ink drops fired out of cartridge.

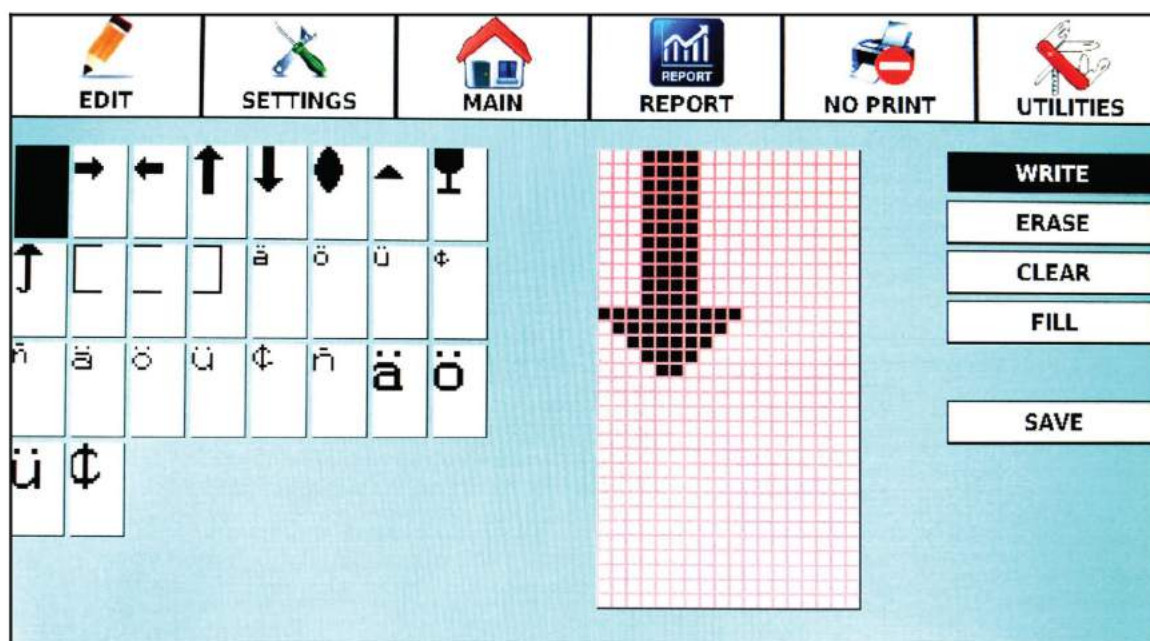
CREDIT: This value represents remaining ink /credits inside a particular cartridge.
****data will be valid only, if smart card provided with a particular cartridge is used.**

FULL CREDIT: this value represents Initial credit/ink drops for fresh cartridge.

WORKING HOURS: This value represents number of working hours for a particular terminal.

4.6 SCREEN: LOGO EDIT

From this screen it is possible to visualize and modify the logos stored in the controller.



The system manages 26 logos of 32x18 dots each, identified by the capital letters from A to Z.

A default logo library is provided for both the print-head and the terminal.

The modified logos are stored in the print head and in the terminal library.

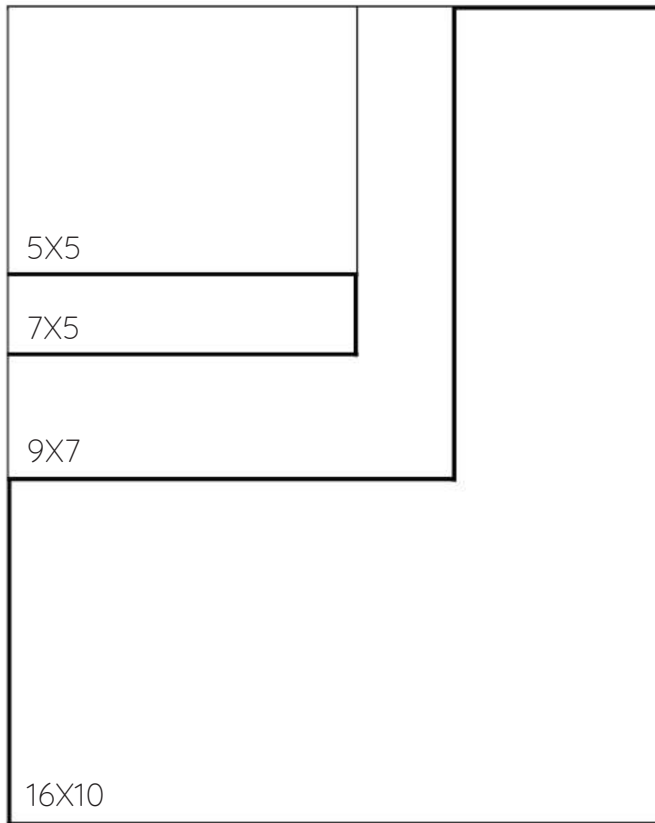
Logos can be used to create graphical symbols or characters not available on the keyboard (e.g.: ò, è, à, ±, «, etc.).

Logos can be used inside an hypertext at the same manner of the characters.

Size of logos depends on the last font matrix selected: depending on the currently selected font the logo can be printed as a whole or only partially.

In particular:

- If the 5x5 font is selected, the system will print only the top left part (5 horizontal dots and 5 vertical ones).
- If the 7x5 font is selected, the system will print only the top left part (5 horizontal dots and 7 vertical ones).
- If the 9x7 font is selected, the system will print only the top left part (7 horizontal dots and 9 vertical ones).
- If the 16x10 font is selected, the system will print all the 16x10 logo etc.



- If more than one logo is printed one after the other, they will be printed together to form a continuous image.
- If they are used separately, they will be printed with an empty column in between each one.
- The latter is useful when logos are used to create characters that are not available on the keyboard.
- Logos and characters can be used together. For example a rectangle can be created and text placed inside of it.
- There is a special character to define the start of a logo once inserted into the hyper text, logos are selected using the A-Z letters.
- The logo sequence must end with the special character signalling logo end.
- Press **WRITE** to draw and use the pen on the touch screen. To stop drawing or writing press **WRITE** or **ERASE** again.
- To erase press **ERASE** then move the pen on the touch screen. When you want to stop deleting press **ERASE** again or **WRITE**.

5. SYSTEM MAINTENANCE AND SERVICE

5

The system does not require any complex maintenance operations.

Attention: use only solvents and inks which are approved by the manufacturer.

CLEANING THE SCREEN OF THE TERMINAL

Use alcohol or light detergents to remove dirt or traces of ink.
Rub with a soft cloth.

CLEANING THE PRINT HEAD

Use either paper or a clean cloth which leaves no threads to remove dirt or traces of dried ink from the print head.

First spray the cloth with solvent before using it on the affected areas

If the nozzle plate needs to be cleaned, avoid moving dirt from one nozzle to another by using a cloth to clean transversally to the row of nozzles.

Attention: do not directly wash the nozzle plate with solvent.

PURGING THE PRINT HEAD

This operation should not be performed while the system is printing.
Touch the purge button which is located on the terminal until all of the nozzles start firing.
Only a small amount of ink will be expelled.

CLEANING THE PRINthead

The printhead is open to allow the cartridge to be inserted.
In a dusty environment, the internal parts have to be periodically cleaned to avoid that any debris, especially corrosive ones, might damage electronic internals or prevents proper cooling.
Use compressed air to gently remove any dirt. Carefully operate to avoid any damage.

REPLACING THE CARTRIDGE

This operation is very simple.
Lower the lever and pull up the old cartridge.
Insert a new one and raise the lever.

Once inserted, insert also the smart card which came together with the new cartridge.
After a few seconds the full ink credit will be shown on the start-up page.

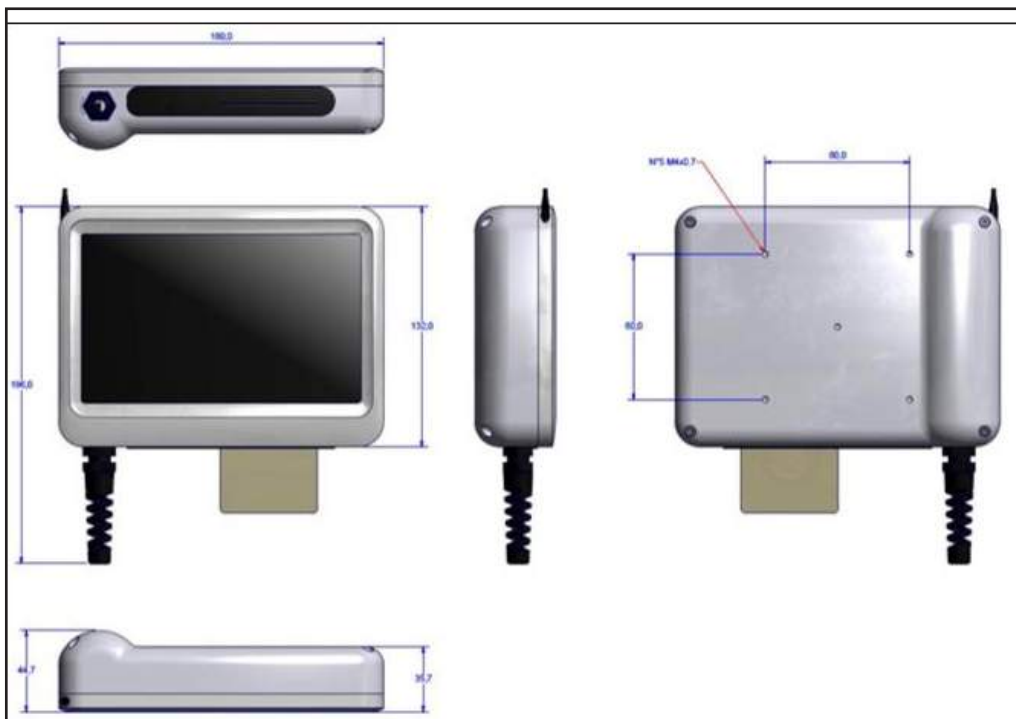
REPLACING FILTERS

No filters are present in this system.

6. TECHNICAL FEATURES

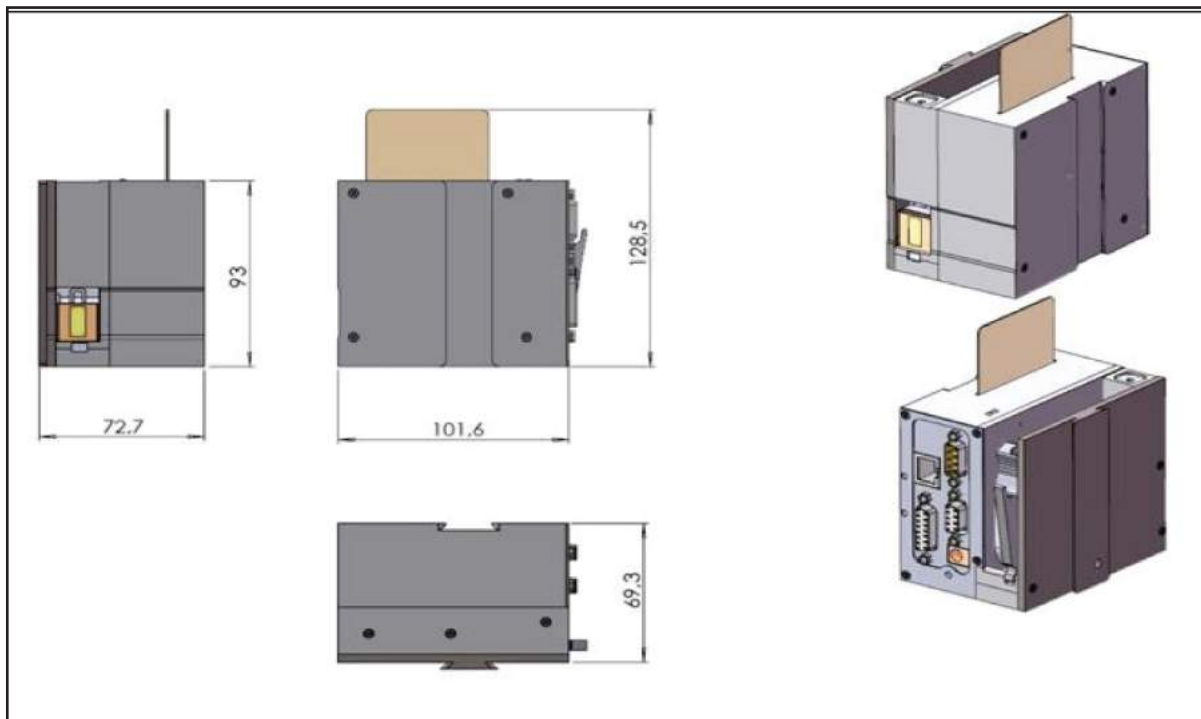
6.1 TERMINAL WITH TOUCH SCREEN

- QWERTY keyboard on the touch screen when required
- 7" LCD TFT colour display with backlight, resolution 800x480
- 24Mz microprocessor
- Real Time Clock with battery
- Buzzer
- Support for stylus
- The terminal can manage more than one printhead
- Icon Touch operator interface
- USB 2.0 port
- Smart card reader
- Spring cable connection to printhead, length 1.5 meter when extended
- IP64 degree of protection
- Power supply: 15Vdc
- Power consumption: 0.2A @ 15Vdc
- Operating temperature: 0-50 °C



6.2 PRINTHEAD FOR 230 ML CARTRIDGES

- 25Mz microprocessor
- Maximum print height **12.7 mm**
- Maximum vertical resolution **600 mm**
- Maximum print speed **60 mpm**
- Integrated product sensor (photocell)
- PNP/NPN external product sensor (photocell) connection
- PNP speed sensor (encoder) connection
- PNP/NPN BCD (Binary Coded Decimal) signal connection (4 signal + strobe)
- Alarm relay. NO and NC free voltage contacts
- RJ45 shielded plug for Ethernet 10/100 Mbps
- Spring- loaded adapter for continued maintenance of a minimum distance between print head and substrate (optional)
- Smart card reader
- ON and Alarm LED's
- Power supply: 15Vdc
- Power consumption: 0.1A @ 15Vdc
- Operating temperature: 0-50 °C



6.3 POWER SUPPLY

- Input voltage: 90-264 Vac, 47-63Hz, 135-370 Vdc, 0.5 A
- Output voltage: 15Vdc, 1.5A (25 W)
- 3 pole AC inlet IEC320-14
- Protections: short circuit, overload, over- voltage
- Operating temperature: 0-50 °C
- Efficiency: 80%
- Approvals: UL, CUL, TUV, CB, FCC, CE
- Cable to the printhead: 2 meters
- Dimensions: 58 x 34 x 100 mm



6.4 SOFTWARE AND PRINT FEATURES

- REPEAT PRINT with selectable time from 1 to 65535, steps of 1ms
- PRINT DELAY with selectable time from 1 to 65535, steps of 1ms
- INTER-COLUMN SPACING selectable from 1 to 65535, steps of 1μs
- Five COUNTERS are available with large flexibility: incremental / decremental up to 9 digits, with / without zeros, numerical (0-9) or alphabetical (A-Z), selectable counting step, standard and batch (they vary only when a defined quantity has been reached)
- Up to 200 HYPERTEXTS can be stored into the terminal
- HYPERTEXT editing and selection can be achieved through Ethernet or through BCD input lines
- 26 customizable LOGOS
- Five font matrixes: 5x5, 7x5, 9x7, 15x10, 15x10, 32x18 dots
- Each HYPERTEXT is associated with its related print report to aid production management
- Date (day, month, year) and time codes (hour, minutes) in any sequence (dd-mm-yy, mm-dd-yy, etc.) and format (1, 2, 4 year digits; month in digits or letters based on the selected language)
- Expiration date in any sequence and format
- Julian date (1-365) and week (1-52) print
- PRINTED SHIFT CODES. Up to 4 shifts can be scheduled associating an identifying code
- BATCH CODE print
- Bold, reverse and inverted print
- BCD input lines
- SHAFT ENCODER management
- PRINT SPEED: up to 60 m / min
- Max image length (no bold): about 800 mm
- Number of text lines: up to five with a 5x5 font matrix
- Different font matrixes can be printed in the same image
- Available barcodes: EAN 13, CODE 39, UPC-A, ITF (Interleaved 2/5)

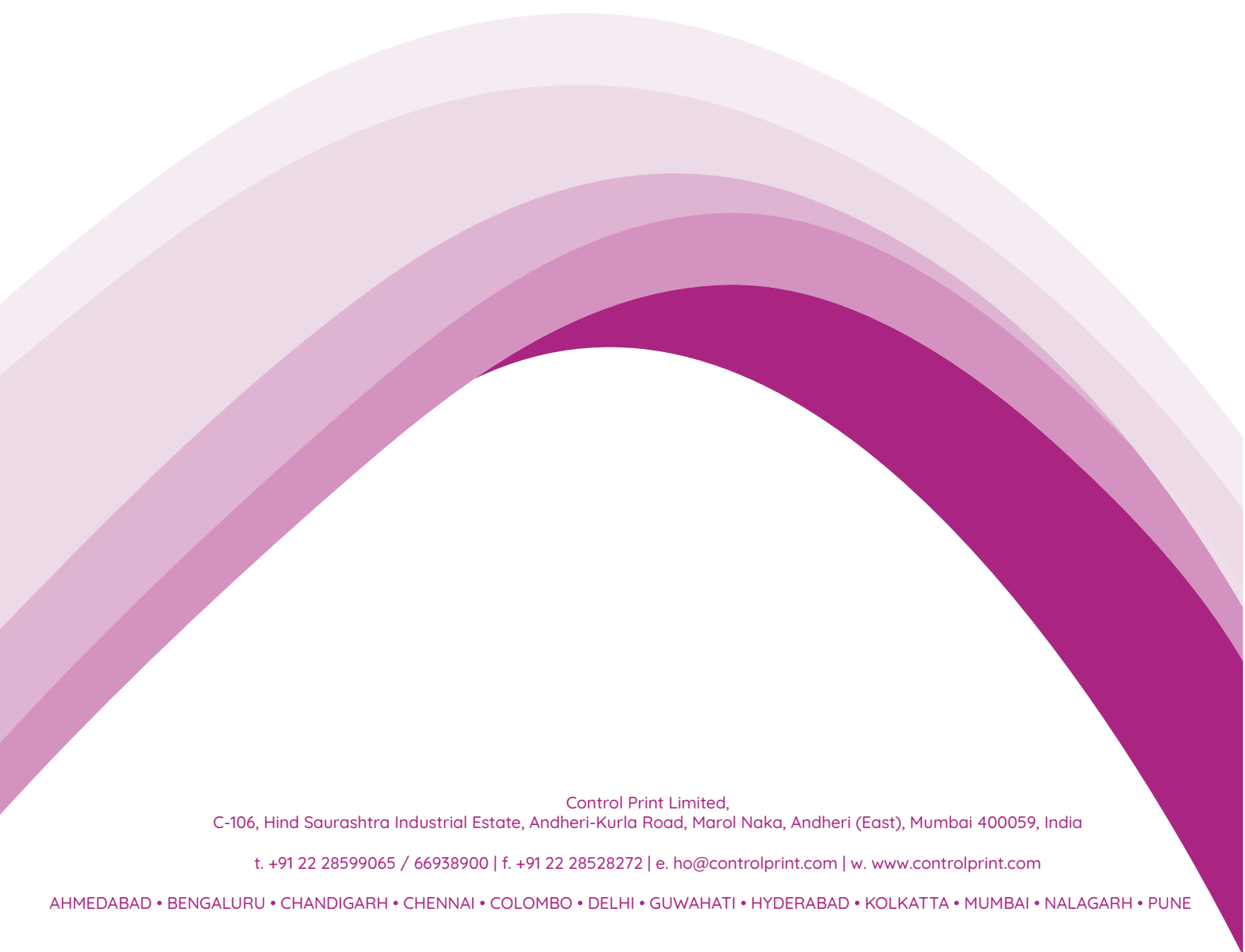
7. PART CODES

7.1 SPARE PART NUMBERS

TIJ GODAVARI SPARE PART CODE LIST		
Description	SAP Part Number	UOM
CP GODAVARI TIJ PRINTER	30270044	01 Nos
SENSOR OBT GODAVARI TIJ/ NARMADA	20370024	Parts included in above code (Full printer set)
TSC CONTROLLER GODAVARI TIJ ASSY	20270068	
PRINT HEAD GODAVARI TIJ ASSY	20270070	
7" TERMINAL BRACKET GODAVARI TIJ	10200826	
TIJ POWER SUPPLY 15V 1.5A EV2	20270010	
TIJ BRACKETORY ASSY WITH FINE ADJ. GODAVARI/NARMADA	20270018	
ENCODER ASSY GODAVARI TIJ/NARMADA 2500 PPR / 5000 PPR	20370027/ 20230013	Optional 01 Nos
SENSOR COLOR MARK GODAVARI TIJ/ NARMADA	20370026	Optional 01 Nos
TOWER LIGHT ASSY GODAVARI TIJ/ NARMADA	20370030	Optional 01 Nos
GAP SENSOR GODAVARI TIJ/ NARMADA	20370025	Optional 01 Nos
PRINT HEAD GODAVARI TIJ/ NARMADA	20270070	Optional for add on (Dual head purpose) 01 Nos
INTERCONNECT CABLE TIJ MINI / MAXI	20370031	Optional for add on (Dual head purpose) 01 Nos

7.2 CONSUMABLE'S PART CODE

Consumables		
Description	SAP Part Number	UOM
TIJ NP 001 BLACK INK 45ML CARTRIDGE(SC)TIJ	50320005	Nos
TIJ NP 002 BLACK INK 45ML CARTRIDGE(SC)	50310200	Nos
TIJ AQ 001 BLACK INK CARTRIDGE, 45ML(SC)	30360017	Nos



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